MTH401 - Differential Equations (Quiz-3)

Question # 4 of 10 (Start time: 01:12:16 PM, 05 January 2026)

The flow of current is _____ proportional to the resistance.

Select the correct option

- (●) inversely
- ( ) directly

MTH401 - Differential Equations (Quiz-3) Qu

Question # 3 of 10 (Start time: 01:12:02 PM, 05 January 2026)

If the system is impressed upon by a _____ force and there is no damping force then there is no transient term in the solution.

Select the correct option

○	periodic
○	friction
○	gravitational
◉	applied

MTH401 - Differential Equations (Quiz-3)

Question # 2 of 10 (Start time: 01:11:49 PM, 05 January 2026)

Vibration of an object on equilibrium point is called simple harmonic motion when the restoring force is proportional to _____.

Select the correct option

○	Mass
◉	Displacement
○	Time
○	A spring constant

MTH401 - Differential Equations (Quiz-3)

Question # 1 of 10 (Start time: 01:11:29 PM, 05 January 2026)

To reduce any Cauchy –Euler differential equation into a differential equation with _____ coefficients we often use substitution

$$x = e^t$$

Select the correct option

◉	constant
○	variable

Clic

Question # 10 of 10 (Start time: 01:09:52 PM, 05 January 2026)

We can derive a differential equation governing the motion of a mass attached to spring when the Newton's second law combined with _____.

Select the correct option

- Newton's 3rd law
- Hook's law

Click to

MTH401 - Differential Equations (Quiz-3)

Question # 9 of 10 (Start time: 01:09:33 PM, 05 January 2026)

A power series in (x-2) is an infinite series of the form

$$\sum_{n=0}^{\infty} c_n(x-2)^n = c_0 + c_1(x-2) + c_2(x-2)^2 + \dots$$

the number 2 is called _____.

Select the correct option

○	Base of power series
○	Radius of power series
●	Center of power series
○	none of these

MTH401 - Differential Equations (Quiz-3)

Question # 7 of 10 (Start time: 01:08:58 PM, 05 January 2026)

If

$$\lambda^2 - \omega^2 = 0$$

and

$$\beta = k$$

then the system is said to be _____ damped.

Select the correct option

◉	critically
○	over
○	under
○	none of these

MTH401 - Differential Equations (Quiz-3)

Question # 6 of 10 (Start time: 01:08:40 PM, 05 January 2026)

The power series, $1 + x + \frac{x}{2!} + \frac{x^3}{3!} + \ldots = _____$.

Select the correct option

○	$\sin x$
○	$\ln(1+x)$
◉	e^x
○	$\cos x$

MTH401 - Differential Equations (Quiz-3)

Question # 8 of 10 (Start time: 01:09:17 PM, 05 January 2026)

If E(t)=0, R=0 (E(t) is the source voltage & R is the resistance) then the electric vibration can be called free _____.

Select the correct option

○	damped oscillation
◉	un-damped oscillation

MTH401 - Differential Equations (Quiz-3)

Question # 5 of 10 (Start time: 01:08:23 PM, 05 January 2026)

The quantity $Z = \sqrt{X^2 + R^2}$ is called _____ of the circuit.

Select the correct option

- (●) impedance
- ( ) reactance

MTH401 - Differential Equations (Quiz-3)

Question # 4 of 10 (Start time: 01:06:05 PM, 05 January 2026)

The D.E

$$r\frac{d^2u}{dr^2} + 2\frac{du}{dr} = 0$$

where the variable

$$r > 0$$

represents the radial distance measured _____ from the center of the sphere.

Select the correct option

◉	outward
○	inward
○	center
○	origin

MTH401 - Differential Equations (Quiz-3)

Question # 3 of 10 (Start time: 01:07:48 PM, 05 January 2026)

The velocity of a particle moving with simple harmonic motion is ---------- at the mean position.

Select the correct option

◉	Maximum
○	Minimum
○	Zero
○	None of these

MTH401 - Differential Equations (Quiz-3)

Question # 2 of 10 (Start time: 01:07:32 PM, 05 January 2026)

For solving non homogeneous differential equation, we use either the method of _____.

Select the correct option

◉	a&b both
○	Undetermined coefficient
○	None of them
○	Variation of parameter

MTH401 - Differential Equations (Quiz-3)

Question # 1 of 10 (Start time: 01:07:13 PM, 05 January 2026)

The displacement measured below the equilibrium position are _____.

Select the correct option

- (●) negative
- ( ) positive

MTH401 - Differential Equations (Quiz-3)

Question # 10 of 10 (Start time: 01:13:39 PM, 05 January 2026)

How are frequency and period related in simple harmonic motion?

Select the correct option

○	They are directly related
○	None of the above
○	Their sum is constant
◉	They are inversely related

MTH401 - Differential Equations (Quiz-3)

Question # 9 of 10 (Start time: 01:13:26 PM, 05 January 2026)

A _____is an electrical component that limits or regulates the flow of electrical current in an electrical circuit.

Select the correct option

○	Inductor
◉	resistor
○	voltage
○	capacitor

MTH401 - Differential Equations (Quiz-3)

Question # 8 of 10 (Start time: 01:13:15 PM, 05 January 2026)

Consider a power series

$$\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^2} x^n = x - \frac{x^2}{2^2} + \frac{x^3}{3^2} + \dots.$$

, then the center of series is _____.

Select the correct option

○	1
○	x
○	x^2
◉	0

MTH401 - Differential Equations (Quiz-3)

Question # 7 of 10 (Start time: 01:13:02 PM, 05 January 2026)

The quantity $Z = \sqrt{X^2 + R^2}$ is called _____ of the circuit.

Select the correct option

- (●) impedance
- ( ) reactance

MTH401 - Differential Equations (Quiz-3)

Question # 6 of 10 (Start time: 01:12:50 PM, 05 January 2026)

Ordinary differential equation $\left(\frac{dy}{dx}\right)^3 + \frac{d^2y}{dx^2} + y = 9$, is of order ------.

Select the correct option

○	2
○	1
◉	3
○	0

MTH401 - Differential Equations (Quiz-3)

Question # 5 of 10 (Start time: 01:12:38 PM, 05 January 2026)

The gravitational force exerted by the earth on a body of mass m is called ---------- of the body.

Select the correct option

- (●) weight
- ( ) Force